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Dept. -: School of Computing and Engineering

Advanced Linux BYOD 1 Practical Examination

Linux Workspace Investigation and Data Preparation

Maximum Marks: 100
Mode: Bring Your Own Device (BYOD)
Time: 2 Hours

General Instructions

Use only Linux terminal commands learned in class.

Do not use GUI tools.

All work must be demonstrated.

Preserve command history.

You must explain your actions when asked.

Do not modify files unless instructed.

Internet usage is prohibited.

Practical Scenario

You are working as a **Linux Operations Assistant** in a company called **DataServe Solutions**.

A former intern has left behind a messy project workspace that must be:

✓ inspected
✓ cleaned
✓ analysed
✓ organised
✓ reported
✓ archived for delivery

The workspace contains:

- duplicate user data
- system logs with errors
- temporary files
- empty files
- large files consuming disk space
- fragmented message files
- incomplete reports

Management has instructed you to investigate and prepare a clean deliverable package.

Provided Workspace Structure

After extraction, you will see:

```
project_workspace/
│
├── users/
│   ├── users.txt
│   ├── new_users.txt
│   └── duplicate_users.txt
│
├── logs/
│   ├── system.log
│   ├── access.log
│   └── old.log
│
├── messages/
│   ├── message_part_01.txt
│   ├── message_part_02.txt
│   ├── message_part_03.txt
│
├── temp/
│   ├── empty1.txt
│   ├── empty2.txt
│   └── unused.tmp
│
├── reports/
│   ├── report_old.txt
│   └── draft.txt
│
├── data/
│   ├── large_data.bin
│   └── sample.txt
│
└── README.txt
```

File Contents

users/users.txt

```
Rimmi
Mehak
Tarunesh
Murli
Krishna
```

```
Prince
Keshav
Archishman
Prince
Krishna
Mehak
```

---

logs/system.log

```
INFO system started
ERROR disk failure
INFO backup running
ERROR network unreachable
INFO service restarted
ERROR permission denied
```

---

logs/access.log

```
login success
login failed
login success
login success
login failed
```

---

messages/message_part_01.txt

```
System maintenance
```

messages/message_part_02.txt

```
completed successfully
```

messages/message_part_03.txt

```
today at 5 PM
```

---

data/sample.txt

```
linux operations practice file
this file is used for counting and viewing
```

---

temp files

(empty)

Your Mission

Investigate the workspace and prepare a **clean, organised, production-ready deliverable**.

Part A — Workspace Exploration (15 Marks)

1. Extract the workspace archive.

```
mehak@DESKTOP-LALCM14:~$ unzip "project_workspace (1).zip"
Archive:  project_workspace (1).zip
   creating: project_workspace/users/
   creating: project_workspace/logs/
   creating: project_workspace/messages/
   creating: project_workspace/temp/
   creating: project_workspace/reports/
   creating: project_workspace/data/
  inflating: project_workspace/README.txt
  inflating: project_workspace/users/users.txt
  inflating: project_workspace/users/new_users.txt
  inflating: project_workspace/users/duplicate_users.txt
  inflating: project_workspace/logs/system.log
  inflating: project_workspace/logs/access.log
  inflating: project_workspace/logs/old.log
  inflating: project_workspace/messages/message_part_01.txt
  inflating: project_workspace/messages/message_part_02.txt
  inflating: project_workspace/messages/message_part_03.txt
  inflating: project_workspace/temp/empty1.txt
  inflating: project_workspace/temp/empty2.txt
  inflating: project_workspace/temp/unused.tmp
  inflating: project_workspace/reports/report_old.txt
  inflating: project_workspace/reports/draft.txt
  inflating: project_workspace/data/large_data.bin
  inflating: project_workspace/data/sample.txt
```

2. Navigate into the workspace.

```
mehak@DESKTOP-LALCM14:~$ cd project_workspace
```

3. Display your current directory.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ pwd
/home/mehak/project_workspace
```

4. List all files including hidden ones.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ ls -la
total 36
drwxr-xr-x  8 mehak mehak 4096 Feb 20 05:26 .
drwxr-x--- 34 mehak mehak 4096 Feb 20 05:26 ..
-rw-r--r--  1 mehak mehak   35 Feb 14 08:49 README.txt
drwxr-xr-x  2 mehak mehak 4096 Feb 14 08:49 data
drwxr-xr-x  2 mehak mehak 4096 Feb 14 08:49 logs
drwxr-xr-x  2 mehak mehak 4096 Feb 14 08:49 messages
drwxr-xr-x  2 mehak mehak 4096 Feb 14 08:49 reports
drwxr-xr-x  2 mehak mehak 4096 Feb 14 08:49 temp
drwxr-xr-x  2 mehak mehak 4096 Feb 14 08:49 users
```

5. Display disk usage of each top-level folder.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ du -sh *
4.0K    README.txt
1.1M    data
16K     logs
16K     messages
12K     reports
8.0K    temp
8.0K    users
```

6. Identify which folder consumes the most space.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ du -sh *
4.0K    README.txt
1.1M    data
16K     logs
16K     messages
12K     reports
8.0K    temp
8.0K    users
```

7. Identify file types present in workspace.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ file */* README.txt
data/large_data.bin:          data
data/sample.txt:              ASCII text
logs/access.log:              ASCII text
logs/old.log:                 ASCII text, with no line terminators
logs/system.log:              ASCII text
messages/message_part_01.txt: ASCII text, with no line terminators
messages/message_part_02.txt: ASCII text, with no line terminators
messages/message_part_03.txt: ASCII text, with no line terminators
reports/draft.txt:            ASCII text, with no line terminators
reports/report_old.txt:       ASCII text, with no line terminators
temp/empty1.txt:              empty
temp/empty2.txt:              empty
temp/unused.tmp:              ASCII text, with no line terminators
users/duplicate_users.txt:    empty
users/new_users.txt:          empty
users/users.txt:              ASCII text
README.txt:                   ASCII text, with no line terminators
mehak@DESKTOP-LALCM14:~/project_workspace$
```

Part B — User Data Processing (15 Marks)

1. Display users.txt in sorted order.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ sort users/users.txt
Archishman
Keshav
Krishna
Krishna
Mehak
Mehak
Murli
Prince
Prince
Rimmi
Tarunesh
```

2. Create a file containing only unique users.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ sort users/users.txt | uniq > users/unique_users.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ cat users/unique_users.txt
Archishman
Keshav
Krishna
Mehak
Murli
Prince
Rimmi
Tarunesh
```

3. Create a file containing duplicate users.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ sort users/users.txt | uniq -d > users/duplicate_users_found.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ cat users/duplicate_users_found.txt
Krishna
Mehak
Prince
```

4. Count total number of users.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ wc -l users/users.txt
11 users/users.txt
```

5. Reverse each username and store result in a new file.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ rev users/users.txt > users/reversed_users.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ cat users/reversed_users.txt
immiR
kaheM
hsenuraT
ilruM
anhsirK
ecnirP
vahseK
namhsihcrA
ecnirP
anhsirK
kaheM
```

Part C — Log Investigation (15 Marks)

1. Extract only ERROR lines from system.log.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ grep "ERROR" logs/system.log
ERROR disk failure
ERROR network unreachable
ERROR permission denied
```

2. Count number of errors.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ grep -c "ERROR" logs/system.log
3
```

3. Extract only “login failed” entries from access.log.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ grep "login failed" logs/access.log
login failed
login failed
```

4. Sort log entries.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ sort logs/system.log
ERROR disk failure
ERROR network unreachable
ERROR permission denied
INFO backup running
INFO service restarted
INFO system started
```

5. Save processed results into error_report.txt.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ echo "---- ERROR REPORT ----" > logs/error_report.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ grep "ERROR" logs/system.log >> logs/error_report.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ echo "Total Errors: $(grep -c ERROR logs/system.log)" >> logs/erro
r_report.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ echo "---- Login Failed Attempts ----" >> logs/error_report.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ grep "login failed" logs/access.log >> logs/error_report.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ cat logs/error_report.txt
---- ERROR REPORT ----
ERROR disk failure
ERROR network unreachable
ERROR permission denied
Total Errors: 3
---- Login Failed Attempts ----
login failed
login failed
```

Part D — Message Reconstruction (10 Marks)

1. Identify message parts.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ ls messages/
message_part_01.txt  message_part_02.txt  message_part_03.txt
```

2. Arrange them in correct order.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ cat messages/message_part_01.txt
System maintenancemehak@DESKTOP-LALCM14:~/project_workspace$ cat messages/message_part_02.txt
completed successfullymehak@DESKTOP-LALCM14:~/project_workspace$ cat messages/message_part_03.txt
today at 5 PMmehak@DESKTOP-LALCM14:~/project_workspace$
```

3.
4. Combine into single sentence.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ cat messages/message_part_01.txt \
messages/message_part_02.txt \
messages/message_part_03.txt > processed/final_message.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ cat processed/final_message.txt
System maintenancecompleted successfullytoday at 5 PMmehak@DESKTOP-LALCM14:~/project_workspace$
```

5.
6. Save into:

`processed/final_message.txt`

```
mehak@DESKTOP-LALCM14:~/project_workspace$ cat messages/message_part_01.txt \
messages/message_part_02.txt \
messages/message_part_03.txt > processed/final_message.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ cat processed/final_message.txt
System maintenancecompleted successfullytoday at 5 PMmehak@DESKTOP-LALCM14:~/project_workspace$
```

Part E — Cleanup Operations (15 Marks)

1. Identify empty files.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ find . -type f -empty
./users/new_users.txt
./users/duplicate_users.txt
./temp/empty1.txt
./temp/empty2.txt
```

2. Verify they contain no data.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ wc -c temp/empty1.txt
0 temp/empty1.txt
```

3.

4. Delete only empty files.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ find . -type f -empty -delete
mehak@DESKTOP-LALCM14:~/project_workspace$ find . -type f -empty -delete
```

5. Remove temporary files.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ rm temp/*.tmp
```

6. Verify cleanup success.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ ls temp/
```

Part F — File Viewing and Analysis (10 Marks)

1. View full content of sample.txt.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ cat data/sample.txt
linux operations practice file
this file is used for counting and viewing
```

2.
3. Display first 2 lines.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ head -n 2 data/sample.txt
linux operations practice file
this file is used for counting and viewing
```

4.
5. Display last 1 line.

```
mehak@DESKTOP-LALCM14:~/project_workspace$ tail -n 1 data/sample.txt
this file is used for counting and viewing
```

6.
7. Open using pager and navigate.

8.
```
mehak@DESKTOP-LALCM14:~/project_workspace$ less data/sample.txt
```
9. Count lines, words, characters.
```
mehak@DESKTOP-LALCM14:~/project_workspace$ wc data/sample.txt
 2 12 74 data/sample.txt
```
10.

Part G — Search and Filtering (10 Marks)

1. Find all text files.
```
mehak@DESKTOP-LALCM14:~/project_workspace$ find . -type f -name "*.txt"
./logs/error_report.txt
./reports/report_old.txt
./reports/draft.txt
./messages/message_part_03.txt
./messages/message_part_01.txt
./messages/message_part_02.txt
./users/users.txt
./users/unique_users.txt
./users/reversed_users.txt
./users/duplicate_users_found.txt.txt
./users/duplicate_users_found.txt
./processed/final_message.txt
./README.txt
```
2.
3. Find files modified recently.
```
mehak@DESKTOP-LALCM14:~/project_workspace$ find . -type f -mtime -1
./logs/error_report.txt
./users/unique_users.txt
./users/reversed_users.txt
./users/duplicate_users_found.txt.txt
./users/duplicate_users_found.txt
./processed/final_message.txt
```
4.
5. Search for word “ERROR” across logs.
```
mehak@DESKTOP-LALCM14:~/project_workspace$ grep "ERROR" logs/*
logs/error_report.txt:---- ERROR REPORT ----
logs/error_report.txt:ERROR disk failure
logs/error_report.txt:ERROR network unreachable
logs/error_report.txt:ERROR permission denied
logs/system.log:ERROR disk failure
logs/system.log:ERROR network unreachable
logs/system.log:ERROR permission denied
```
6.
7. Save search results into a file.

8.
```
mehak@DESKTOP-LALCM14:~/project_workspace$ grep "ERROR" logs/* > logs/search_results.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ cat logs/search_results.txt
logs/error_report.txt:---- ERROR REPORT ----
logs/error_report.txt:ERROR disk failure
logs/error_report.txt:ERROR network unreachable
logs/error_report.txt:ERROR permission denied
logs/system.log:ERROR disk failure
logs/system.log:ERROR network unreachable
```
9. Count number of matching lines using pipe.
10.
```
mehak@DESKTOP-LALCM14:~/project_workspace$ grep "ERROR" logs/* | wc -l
14
```

Part H — Redirection and Reporting (5 Marks)

1. Redirect disk usage output into report file.
2.
```
mehak@DESKTOP-LALCM14:~/project_workspace$ du -sh * > reports/statistics_report.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ cat reports/statistics_report.txt
4.0K    README.txt
1.1M    data
24K     logs
16K     messages
8.0K    processed
16K     reports
4.0K    temp
24K     users
```
3. Append current date and time.
4.
```
mehak@DESKTOP-LALCM14:~/project_workspace$ date >> reports/statistics_report.txt
mehak@DESKTOP-LALCM14:~/project_workspace$ cat reports/statistics_report.txt
4.0K    README.txt
1.1M    data
24K     logs
16K     messages
8.0K    processed
16K     reports
4.0K    temp
24K     users
```
5. Save system summary message using tee.
6.
```
mehak@DESKTOP-LALCM14:~/project_workspace$ echo "Workspace investigation completed successfully." | tee repor
ts/system_summary.txt
Workspace investigation completed successfully.
```

Part I — Final Packaging (5 Marks)

Create final folder:

```
deliverable/
```

Place inside:

- unique users file
- duplicate users file
- error report
- reconstructed message
- statistics report

```
mehak@DESKTOP-LALCM14:~/project_workspace$ mkdir deliverable
mehak@DESKTOP-LALCM14:~/project_workspace$ ls
README.txt  data  deliverable  logs  messages  processed  reports  temp  users
mehak@DESKTOP-LALCM14:~/project_workspace$ cp users/unique_users.txt deliverable/
mehak@DESKTOP-LALCM14:~/project_workspace$ cp users/duplicate_users_found.txt deliverable/
mehak@DESKTOP-LALCM14:~/project_workspace$ cp logs/error_report.txt deliverable/
mehak@DESKTOP-LALCM14:~/project_workspace$ cp processed/final_message.txt deliverable/
mehak@DESKTOP-LALCM14:~/project_workspace$ cp reports/statistics_report.txt deliverable/
mehak@DESKTOP-LALCM14:~/project_workspace$ ls deliverable/
duplicate_users_found.txt  error_report.txt  final_message.txt  statistics_report.txt  unique_users.txt
mehak@DESKTOP-LALCM14:~/project_workspace$
```